

PROJECT ABSTRACT

Project Title:

GramDorpon: A Digital Twin Framework for Real-Time Smart Village Management and Disaster Resilience in Bangladesh

1. Problem Statement:

Bangladesh is a country of villages. Despite rapid digitization, there is a massive gap in real-time, integrated data for rural areas. Local administrations (Union Parishads), farmers, and disaster management authorities operate blindly without a unified view of their village's current status.

Currently, solutions are fragmented: one app gives weather, another detects crop diseases, and Google Maps shows outdated satellite images. There is no single platform that visually simulates the real-time condition of a village—specifically water levels (flood risk), crop health, road conditions, and population density. This lack of a "Digital Twin" leads to delayed disaster response, unnecessary crop loss, and poor infrastructure planning.

2. Proposed Solution:

GramDorpon (Bengali for "গ্রামদর্পণ") is a pioneering 3D Interactive Digital Twin Platform. It creates a fully simulated, real-time virtual replica of a Bangladeshi village.

Instead of just showing raw data, GramDorpon visualizes the village in 3D (like a video game). It ingests live data from satellites, weather APIs, and crowdsourced inputs to dynamically change the virtual environment. For example, if a heavy rain simulation is triggered, the virtual water level rises in the 3D scene, and the dashboard automatically issues a "Flood Alert" with actionable advice. It bridges the gap between complex data and common villagers by turning information into a visual, easy-to-understand simulation.

3. Objectives:

- To develop a centralized 3D visualization platform for monitoring rural village infrastructure (roads, water bodies, housing).
- To implement an AI-driven predictive alert system for natural calamities (specifically floods and crop diseases).
- To provide local authorities (Union Parishads) with a data-driven dashboard for prioritizing infrastructure maintenance (e.g., which road to fix first).
- To empower farmers with real-time crop health and water table insights to increase yield.

4. Methodology / Technology Used:

The project is built using a modern Full-Stack architecture focusing on high-performance 3D rendering and API integration:

- Frontend (Visualization):** React.js with Vite for fast loading, combined with Three.js and CesiumJS to render the 3D village, houses, trees, and dynamic water levels inside a web browser.
- Backend (Server & Logic):** Node.js with Express.js to handle API requests and simulation logic.

- c. **Data Sources (APIs):** Integration of OpenWeatherMap API (for real-time rainfall/temperature) and Google Earth Engine API (for satellite image processing of crop/water bodies).
- d. **Database:** PostgreSQL to store village demographic and infrastructure data.
- e. **AI/ML:** Python (TensorFlow) for processing satellite images to detect crop health (e.g., identifying Blast disease or waterlogging).
- f. **Hosting:** Scalable on AWS EC2 and S3 cloud servers.

5. Expected Outcomes:

- a. A fully functional, interactive 3D prototype of a selected village.
- b. A prediction model that accurately alerts users 48 hours in advance if water levels are about to cross danger thresholds.
- c. A "Crop Health Map" that identifies stressed agricultural zones 2 weeks earlier than manual inspection.
- d. An analytics dashboard that shows which village roads are most vulnerable to monsoon rains, saving maintenance costs by up to 30%.

6. Impact and Feasibility:

- a. **Impact (Societal):** Bangladesh faces annual floods. By predicting which specific house or crop field will be affected first, GramDorpon can save thousands of livestock and lives. It directly supports the government's "Smart Bangladesh" vision by digitizing rural planning.
- b. **Feasibility (Technical):** The project uses 100% open-source technologies (Three.js, React, Python) and publicly available APIs. The prototype is already built and running on a local server, proving that the concept works. Scaling it to other villages only requires uploading their specific geographical data into the database. It is highly cost-effective compared to installing physical sensors everywhere.

7. Team Information:

- **Team Name: Iostream**

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